

✓ Practical Statement 1 : Bit Plane of Image

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

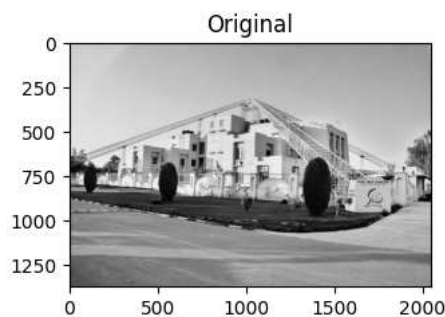
```
img = cv2.imread('nit.jpg', cv2.IMREAD_GRAYSCALE)
```

```
# Create a list to store the 8 bit planes
bit_planes = []
```

```
for i in range(8):
    # Extract the i-th bit:
    # Shift right by 'i' and perform bitwise AND with 1
    plane = (img >> i) & 1
    # Scale to 0-255 for better visibility in display
    bit_planes.append(plane * 255)
```

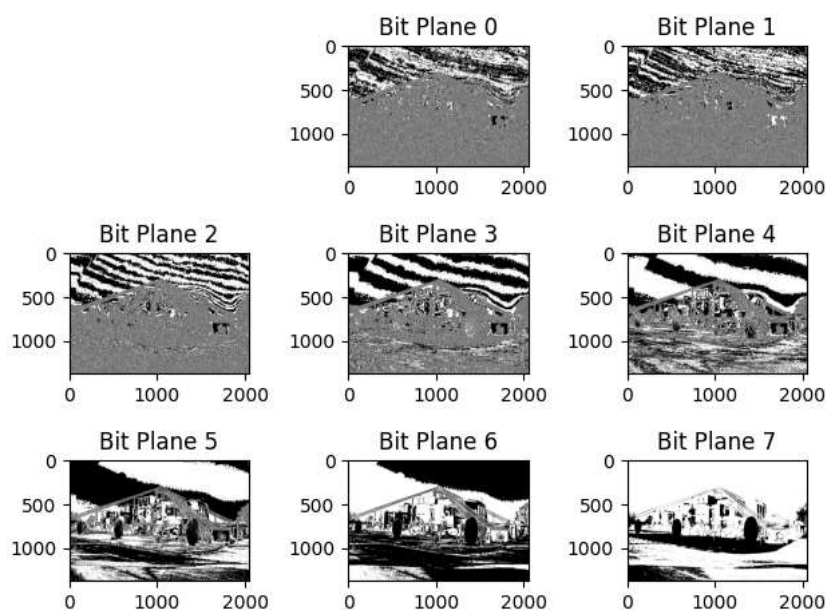
```
# Display the original and the 8 bit planes
plt.figure(figsize=(12, 8))
plt.subplot(3, 3, 1), plt.imshow(img, cmap='gray'), plt.title('Original')
```

```
(<Axes: title={'center': 'Original'}>,
 <matplotlib.image.AxesImage at 0x7b225cbefd40>,
 Text(0.5, 1.0, 'Original'))
```



```
for i in range(8):
    plt.subplot(3, 3, i+2)
    plt.imshow(bit_planes[i], cmap='gray')
    plt.title(f'Bit Plane {i}') # Plane 0 is LSB, Plane 7 is MSB

plt.tight_layout()
plt.show()
```



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✓ Practical Statement 2 : Edge Detection using Sobel Laplasian

```
import cv2
import numpy as np
from matplotlib import pyplot as plt
```

```
img = cv2.imread('img.png', cv2.IMREAD_GRAYSCALE)
```

```
# Sobel Edge Detection (First-order derivative)
# Computes gradients in X and Y directions
sobel_x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)
```

```
sobel_combined = cv2.magnitude(sobel_x, sobel_y)
```

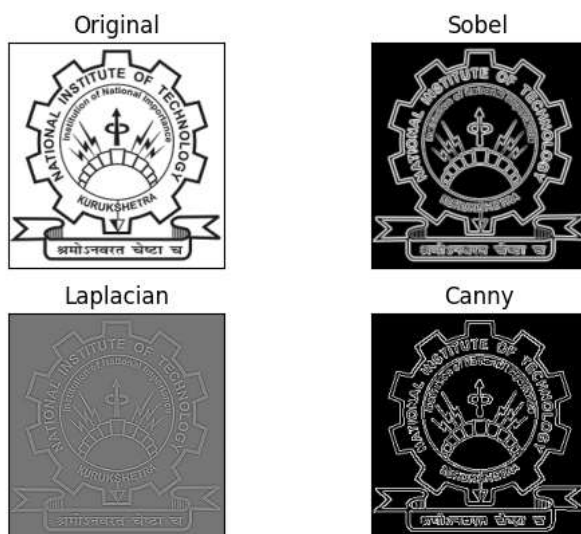
```
# Laplacian Edge Detection (Second-order derivative)
# Detects edges by finding zero-crossings
laplacian = cv2.Laplacian(img, cv2.CV_64F)
```

```
# Canny Optimization (Multi-stage detection)
canny = cv2.Canny(img, 100, 200)
```

```
# Display results
titles = ['Original', 'Sobel', 'Laplacian', 'Canny']
images = [img, sobel_combined, laplacian, canny]
```

```
for i in range(4):
    plt.subplot(2, 2, i+1), plt.imshow(images[i], 'gray')
    plt.title(titles[i])
    plt.xticks([], plt.yticks([]))

plt.show()
```



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